# Node.js and Express

Course summary | Instructor: John Smilga

A short summary of the course: how Node.js works, its modules and npm, and how to build a REST API with Express.

## Week 1-2

### What is Node.js

Node.js is a JavaScript runtime built on the V8 engine that runs outside the browser. It uses an event loop and non-blocking I/O, which suits servers handling many requests. Run a file with node app.js.

### Modules

CommonJS uses require() and module.exports. ES modules use import and export (set "type": "module" in package.json). Each file has its own scope.

### Built-in Modules

Node ships with modules such as fs (files), path, http, os and events. Prefer asynchronous file methods (fs.promises) in servers, and build paths with path.join.

### NPM

npm installs packages: npm init, then npm install name. Dependencies are listed in package.json and stored in node_modules. Use devDependencies for tools and npm run for scripts.

## Week 3-4

### Routing

Express maps an HTTP method and path to a handler: app.get("/users", (req, res) => res.json(users)). Route parameters like /users/:id are read from req.params. Start with app.listen(3000).

### Middleware

Middleware are functions (req, res, next) that run in order. express.json() parses JSON bodies. Call next() to continue. Error handlers take four parameters: (err, req, res, next).

### Building a REST API

Model resources with HTTP verbs: GET reads, POST creates, PUT or PATCH updates and DELETE removes. Return clear status codes such as 200, 201, 400, 404 and 500, with JSON bodies.

### Connecting a Database

Use a driver or an ODM such as Mongoose for MongoDB. Keep the connection string in an environment variable (process.env) and never commit secrets. Use async / await and handle errors.

### Quick reference

| | |
|---|---|
| require("x") | Load a module (CommonJS) |
| app.use(express.json()) | Parse JSON bodies |

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|---|---|
| `res.status(404).json({})` | Send a status and JSON |
| `process.env.PORT` | Read an environment variable |
| `npm run <script>` | Run a package script |